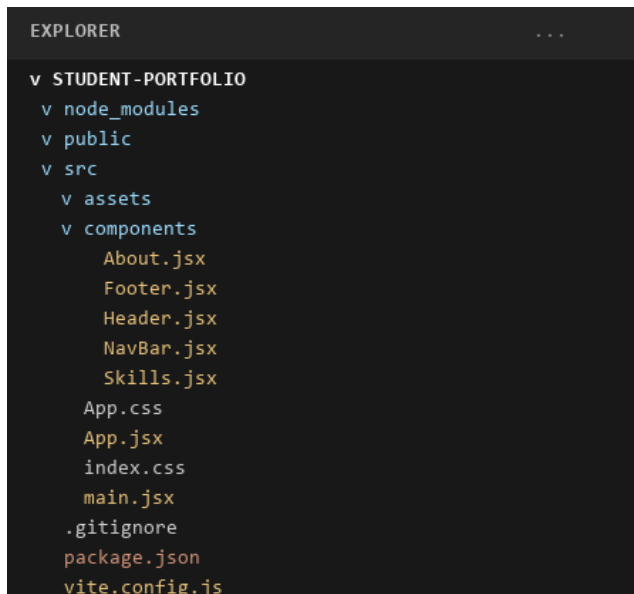


Practical 1: Introduction to React and Component Architecture

Problem Definition:

- Create a React application using Vite for a student portfolio page.
- The application must include at least 4 reusable components: Header, About, Skills, and Footer.
- Each component must be independently structured and composed into a single-page layout.
- No logic or JSX should be duplicated across components.
- Props must be used to pass at least one piece of data into a minimum of 2 components.

Folder Structure



Implementation:

About.jsx

student-portfolio > src > components > About.jsx > About

```
1  import React from 'react';
2  import { UserCheck, GraduationCap, Cpu, Layers } from 'lucide-react';
3  import { resumeData } from '../data/resumeData';
4
5  function About() {
6    const info = resumeData.personalInfo;
7
8    return (
9      <section id="about">
10        <div className="container">
11          <h2 className="section-title">
12            <UserCheck size={26} color="var(--accent-cyan)" /> About Me
13          </h2>
14          <p className="section-subtitle">
15            Professional summary and engineering focus
16          </p>
17
18          <div className="about-grid">
19            <div className="glass-card about-card">
20              <h3 style={{ fontSize: '1.4rem', marginBottom: '1rem', color: 'var(--text-primary)' }}>
21                Professional Summary
22              </h3>
23              <p style={{ color: 'var(--text-secondary)', fontSize: '1.05rem', lineHeight: '1.7' }}>
24                {info.summary}
25              </p>
26            </div>
27
28            <div className="glass-card about-card">
29              // ... (truncated remaining lines)
```

Ln 55, Col 1 Spaces: 2 UTF-8 JavaScript

Footer.jsx

student-portfolio > src > components > Footer.jsx > Footer

```
1  import React from 'react';
2  import { resumeData } from '../data/resumeData';
3
4  function Footer() {
5    const info = resumeData.personalInfo;
6
7    return (
8      <footer className="footer">
9        <div className="container">
10          <p style={{ fontWeight: 700, fontSize: '1rem', color: 'var(--text-primary)' }}>
11            &copy; {new Date().getFullYear()} {info.fullName}
12          </p>
13          <p style={{ color: 'var(--text-muted)', fontSize: '0.88rem', marginTop: '0.35rem' }}>
14            ITUE301: Advanced Web Development Frameworks – Practical 1: Introduction to React & Component Architecture
15          </p>
16          <div style={{ display: 'flex', justifyContent: 'center', gap: '1.5rem', marginTop: '1rem' }}>
17            <a href={info.github} target="_blank" rel="noreferrer" style={{ color: 'var(--accent-cyan)', textDecoration: 'none', fo
18              GitHub {{info.githubUsername}}
19            </a>
20            <a href={info.linkedin} target="_blank" rel="noreferrer" style={{ color: 'var(--accent-cyan)', textDecoration: 'none',
21              LinkedIn
22            </a>
23          // ... (truncated remaining lines)
```

Ln 29, Col 1 Spaces: 2 UTF-8 JavaScript

Header.jsx

student-portfolio > src > components > Header.jsx > Header

```
1  import React from 'react';
2  import { Sparkles, Phone, Mail, MapPin, Download, FileText } from 'lucide-react';
3  import { resumeData } from '../data/resumeData';
4
5  const GithubIcon = () => (
6    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="var(--accent-cyan)" strokeWidth="2" strokeLinecap="round" s
7      <path d="M15 22v-4a4.8 4.8 0 0 1-3.5c3 0 6-2 6-5.5.08-1.25-.27-2.48-1-3.5.28-1.15.28-2.35 0-3.5 0 0-1 0-3 1.5-2.64-.5-5.36-
8      <path d="M9 18c-4.51 2-5-2-7-2"></path>
9    </svg>
10 );
11
12 const LinkedInIcon = () => (
13   <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="var(--accent-cyan)" strokeWidth="2" strokeLinecap="round" s
14     <path d="M16 8a6 6 0 0 1 6 6v7h-4v-7a2 2 0 0 2 2 2 0 0 2 2v7h-4v-7a6 6 0 0 1 6-6z"></path>
15     <rect x="2" y="9" width="4" height="12"></rect>
16     <circle cx="4" cy="4" r="2"></circle>
17   </svg>
18 );
19
20 function Header({ name, tagline, onViewResume }) {
21   const info = resumeData.personalInfo;
22
23   return (
24     <header className="hero-section" id="hero">
25     // ... (truncated remaining lines)
```

Ln 84, Col 1 Spaces: 2 UTF-8 JavaScript

Navbar.jsx

student-portfolio > src > components > NavBar.jsx > NavBar

```
1  import React from 'react';
2  import { Cpu, User, Briefcase, Rocket, GraduationCap, FileText, Download } from 'lucide-react';
3  import { resumeData } from '../data/resumeData';
4
5  function NavBar({ showOnlyResume, setShowOnlyResume }) {
6    return (
7      <nav className="navbar" role="navigation">
8        <div className="container nav-flex">
9          <a
10            href="#hero"
11            className="brand-logo"
12            onClick={(e) => {
13              if (showOnlyResume) setShowOnlyResume(false);
14            }}
15          >
16            <Cpu size={26} color="var(--accent-cyan)" />
17            <span>{resumeData.personalInfo.displayName}</span><span className="gradient-text">.Dev</span></span>
18          </a>
19
20          <ul className="nav-links">
21            {!showOnlyResume ? (
22              <>
23                <li><a href="#about" className="nav-link"><User size={15} /> About</a></li>
24                <li><a href="#skills" className="nav-link"><Cpu size={15} /> Skills</a></li>
25              </>
26            ) : null}
27            // ... (truncated remaining lines)
```

Ln 66, Col 1 Spaces: 2 UTF-8 JavaScript

Skills.jsx

student-portfolio > src > components > Skills.jsx > Skills

```
1  import React from 'react';
2  import { Cpu, Code2, Wrench, CheckCircle2 } from 'lucide-react';
3  import { resumeData } from '../data/resumeData';
4
5  function Skills({ skillList = [] }) {
6    const categorized = resumeData.skills;
7
8    return (
9      <section id="skills">
10        <div className="container">
11          <h2 className="section-title">
12            <Cpu size={26} color="var(--accent-cyan)" /> Skills & Expertise
13          </h2>
14          <p className="section-subtitle">
15            Core technical competencies across hardware, embedded software, and full-stack development
16          </p>
17
18          <div className="glass-card skills-container">
19            /* Categorized Skills Section */
20            <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '2rem' }}>
21
22              <div>
23                // ... (truncated remaining lines)
```

Ln 84, Col 1 Spaces: 2 UTF-8 JavaScript

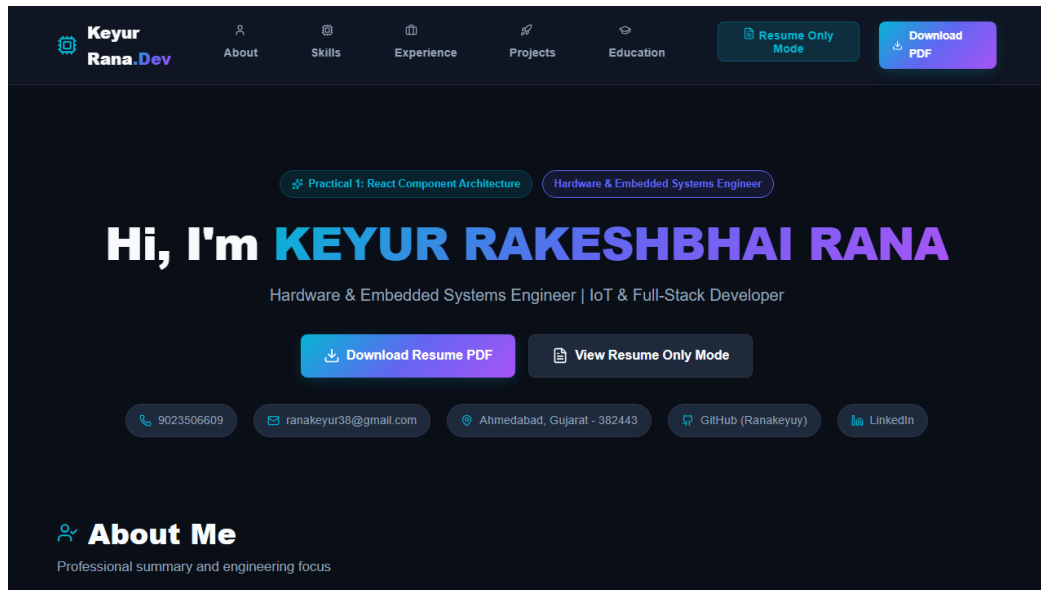
App.jsx

student-portfolio > src > App.jsx > App

```
1  import React, { useState } from 'react';
2  import NavBar from './components/NavBar';
3  import Header from './components/header';
4  import About from './components/about';
5  import Skills from './components/skills';
6  import Experience from './components/experience';
7  import Projects from './components/projects';
8  import Education from './components/education';
9  import ResumeSection from './components/ResumeSection';
10 import Footer from './components/footer';
11 import { resumeData } from './data/resumeData';
12 import './App.css';
13
14 function App() {
15   const [showOnlyResume, setShowOnlyResume] = useState(false);
16
17   const skillsArray = [
18     ...resumeData.skills.languagesAndDomains,
19     ...resumeData.skills.hardwareAndSensors,
20     ...resumeData.skills.toolsAndPlatforms
21   ];
22
23   return (
24     <div className="app-container">
25       <NavBar
26         showOnlyResume={showOnlyResume}
27         setShowOnlyResume={setShowOnlyResume}
28       />
29
30       <main className="main-content">
31         {showOnlyResume ? (
32           // ONLY 2-Page Resume PDF & Download Button View
33           // ... (truncated remaining lines)
```

Ln 64, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



Key Questions:

- What is the role of each component in the overall UI structure?

App → Root component that organizes the entire page.

Navbar → Displays navigation links.

Sidebar → Shows menus or additional options.

Footer → Displays copyright or contact information.

- How does React re-render when props change?

Example :

```
function Child(props) {  
  return <h1>{props.name}</h1>;  
}
```

```
function Parent() {  
  return <Child name="Ranak" />;  
}
```

- **Why is component reusability important in large-scale applications?**

Component reusability means creating a component once and using it multiple times.

Example:

```
<Button text="Login" />  
<Button text="Register" />  
<Button text="Submit" />
```

Instead of writing button code repeatedly, a single Button component is reused.

Advantages:

Less code duplication → Write once, use many times.

Easy maintenance → Fix a bug in one place.

Consistency → Same design and behavior across the application.

Faster development → Developers can reuse existing components.

Better scalability → Large applications become easier to manage.

For example, companies like Facebook, Instagram, and Netflix use thousands of reusable React components to build complex UIs efficiently.

Supplementary Problems:

- Add a NavBar component that highlights the active section.

Implementation:

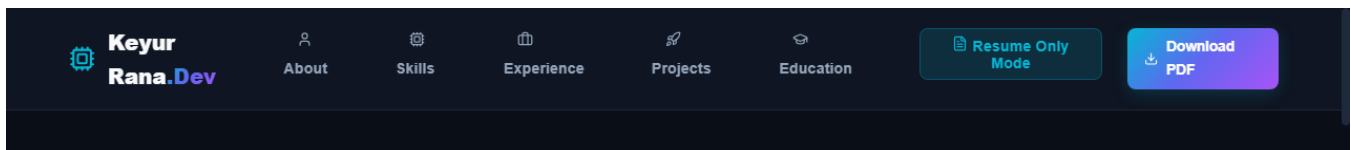
Navbar.jsx

student-portfolio > src > components > NavBar.jsx > NavBar

```
1  import React from 'react';
2  import { Cpu, User, Briefcase, Rocket, GraduationCap, FileText, Download } from 'lucide-react';
3  import { resumeData } from '../data/resumeData';
4
5  function NavBar({ showOnlyResume, setShowOnlyResume }) {
6    return (
7      <nav className="navbar" role="navigation">
8        <div className="container nav-flex">
9          <a
10             href="#hero"
11             className="brand-logo"
12             onClick={(e) => {
13               if (showOnlyResume) setShowOnlyResume(false);
14             }}
15          >
16            <Cpu size={26} color="var(--accent-cyan)" />
17            <span>{resumeData.personalInfo.displayName}<span className="gradient-text">.Dev</span></span>
18          </a>
19
20          <ul className="nav-links">
21            {!showOnlyResume ? (
22              <>
23                <li><a href="#about" className="nav-link"><User size={15} /> About</a></li>
24                <li><a href="#skills" className="nav-link"><Cpu size={15} /> Skills</a></li>
25              </>
26            ) : null}
27          </ul>
28        </div>
29      </nav>
30    );
31  }
```

Ln 66, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



- Pass an array of skills as a prop to the Skills component and render them dynamically.

Implementation:

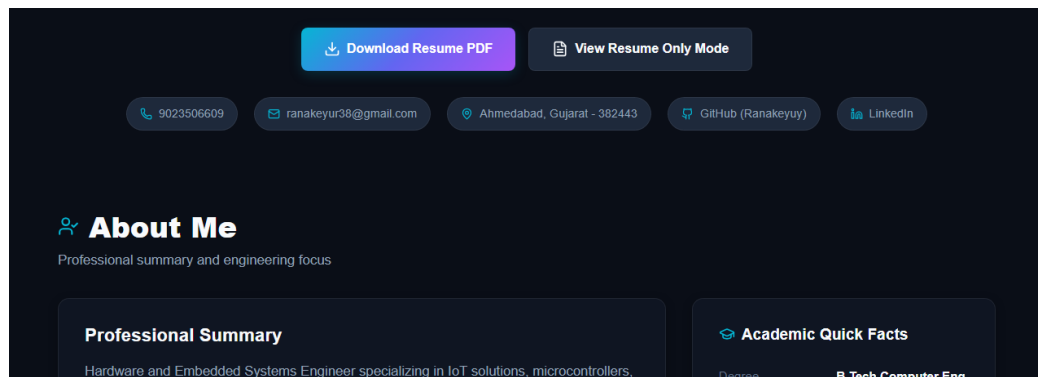
Skills.jsx

student-portfolio > src > components > Skills.jsx > Skills

```
1  import React from 'react';
2  import { Cpu, Code2, Wrench, CheckCircle2 } from 'lucide-react';
3  import { resumeData } from '../data/resumeData';
4
5  function Skills({ skillList = [] }) {
6    const categorized = resumeData.skills;
7
8    return (
9      <section id="skills">
10        <div className="container">
11          <h2 className="section-title">
12            <Cpu size={26} color="var(--accent-cyan)" /> Skills & Expertise
13          </h2>
14          <p className="section-subtitle">
15            Core technical competencies across hardware, embedded software, and full-stack development
16          </p>
17
18          <div className="glass-card skills-container">
19            /* Categorized Skills Section */
20            <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '2rem' }}>
21              <div>
22                // ... (truncated remaining lines)
23            </div>
24          </div>
25        </div>
26      </section>
27    );
28  }
```

Ln 84, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



- Add a theme color prop to the Header and apply it as an inline style.

Implementation:

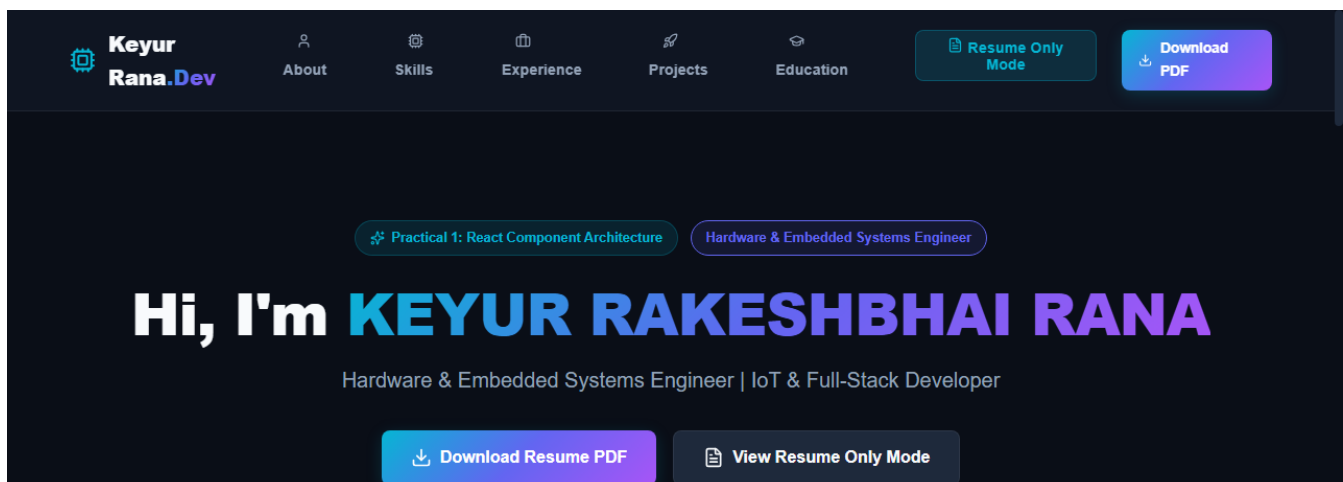
App.jsx

student-portfolio > src > App.jsx > App

```
1  import React, { useState } from 'react';
2  import NavBar from './components/NavBar';
3  import Header from './components/header';
4  import About from './components/about';
5  import Skills from './components/skills';
6  import Experience from './components/experience';
7  import Projects from './components/projects';
8  import Education from './components/education';
9  import ResumeSection from './components/ResumeSection';
10 import Footer from './components/footer';
11 import { resumeData } from './data/resumeData';
12 import './App.css';
13
14 function App() {
15   const [showOnlyResume, setShowOnlyResume] = useState(false);
16
17   const skillsArray = [
18     ...resumeData.skills.languagesAndDomains,
19     ...resumeData.skills.hardwareAndSensors,
20     ...resumeData.skills.toolsAndPlatforms
21   ];
22
23   return (
24     <div className="app-container">
25       <NavBar
26         showOnlyResume={showOnlyResume}
27         setShowOnlyResume={setShowOnlyResume}
28       />
29
30       <main className="main-content">
31         {showOnlyResume ? (
32           // ONLY 2-Page Resume PDF & Download Button View
33           // ... (truncated remaining lines)
```

Ln 64, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



Github Link : <https://github.com/Ranakeyur-31525/awdf>